Ad Hoc Networks 72 (2018) 68–80

Contents lists available at ScienceDirect

Ad Hoc Networks

journal homepage: www.elsevier.com/locate/adhoc

A heterogeneous network selection algorithm based on network attribute and user preference

He-Wei Yu, Biao Zhang*

School of Computer Science and Engineering, South China University of Technology, Guangzhou 510006 Guangdong, China

ARTICLE INFO

Article history:
Received 21 August 2017
Revised 11 January 2018
Accepted 30 January 2018
Available online 31 January 2018

Keywords:
Heterogeneous networks
Network selection
Entropy
FAHP
TOPSIS

ABSTRACT

Mobile terminals are often in the dynamic environment of heterogeneous networks. For some reasons, they need to switch between different networks, namely vertical handover. At this moment, it is a very crucial issue for the mobile terminal to select the best suitable one rapidly from all detected alternative networks on condition that the selection result can avoid the ping-pong effect as much as possible. This paper proposes a heterogeneous network selection algorithm based on the combination of network attribute and user preference. Taking full account of user preferences for each candidate network and the actual situation of heterogeneous networks, the algorithm combines three typical MADM methods, namely FAHP, Entropy and TOPSIS. We first use FAHP to calculate the subjective weights of network attributes and the subjective utility values of all alternatives for four typical traffic classes, and then use Entropy and TOPSIS to respectively get the objective weights of network attributes and the objective utility values of all alternatives. Finally, according to the comprehensive utility value of every candidate network and a threshold, the most appropriate network, whose comprehensive utility value is maximum and greater than the corresponding value of the current network of the mobile terminal, is selected to access. The proposed algorithm not only avoids the one-sided nature of a single algorithm, but also dynamically adjusts the proportion of each algorithm in the final result according to the actual requirements. Simulation results indicate that the proposed algorithm can accurately select the optimal access network, significantly reduce the number of vertical handovers and provide the required QoS and QoE in terms of the quantified benefit from vertical handoff, compared with three existing hybrid algorithms.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Since standards of 4G communication were issued, many research institutions in the world have been researching the next generation of communication standards 5G and relevant technology. For the equilibrium among operators, suppliers and users, various heterogeneous wireless networks (2G, 3G, 4G and WLAN and so on) will coexist and complement for a long time [1,2]. Hence, the trend of heterogeneous networks integration is increasingly evident especially with the advent of Internet of Things [3,4]. How to select a target access point to ensure efficient communication between two terminals or nodes in the system has become a basic requirement for many practical application scenarios, for example Scale-Free Wireless Sensor Networks [5], Social Internet of Vehicles [6,7]. In order to be always best connection (ABC) to obtain a high Quality of Experience (QoE) for users, mobile terminal needs to switch between different networks [8]. This switching behaviour is known as vertical handover. Necessary vertical handovers can bring profit to end users and network operators, while unnecessary vertical handovers can degrade QoE of user and overall network performance [9–11]. Consequently, it has become an interesting and challenging research topic to choose an optimal one from several candidate networks.

Generally, the issue of heterogeneous wireless network selection can be formulated as a problem of multiple attribute decision making (MADM) [12,13]. With respect to attributes considered in this area, there are the following three categories: (1), the QoS characteristics of candidate networks, such as received signal strength (RSS), network load, available bandwidth, data transmission rate, latency, delay jitter, packet loss ratio and bit error rate. (2), the attributes relevant to mobile terminal, such as rate of movement, the angle formed between the direction of movement and the access point (or base station), and remaining power [14]. (3), the user preferences, such as the overall level of personal preference for each network [15].

The typical MADM algorithms available currently are Simple Additive Weighting (SAW), Hierarchical Analysis Process (AHP), Fuzzy Analytic Hierarchy Process (FAHP), Technique for Order Pref-

* Corresponding author.
E-mail addresses: hwyu@scut.edu.cn (H.-W. Yu), z.b21@mail.scut.edu.cn (B. Zhang).

https://doi.org/10.1016/j.adhoc.2018.01.011
1570-8705/© 2018 Elsevier B.V. All rights reserved.

erence by Similarity to an Ideal Solution (TOPSIS), Entropy, Grey Relational Analysis (GRA), Markov Process [16], Weighted Markov Chain (WMC), and VIseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) [17], etc. Each algorithm has its own advantages and disadvantages. Undoubtedly, none of these algorithms can solve all MADM problems perfectly so far. Hence, in the context of heterogeneous network selection, there are two intuitive ways: making improvements on one of the algorithms above (e.g., [18] and [19]) or combining a variety of algorithms so as to get a better comprehensive result. The former only uses a single algorithm and thus can quickly get the final results by some simple operations, but the improvements are very limited. Therefore it has not been used gradually. Whereas the latter uses no less than two kinds of algorithms and thus increases the computation complexity and latency unavoidably, but it can bring together the advantages of the related algorithms, effectively overcome the one sidedness of a single algorithm, and show a strong ability to meet the environment. Hence it is more popular and widely used [20].

It should be noted that in addition to these schemes based on MADM, many intelligent algorithms have also been applied for this issue, such as Artificial Neural Network (ANN), Genetic Algorithm (GA), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Simulated Annealing (SA), Q-Learning [21,22], GA-SA (a hybrid of GA and SA) [23]. However, all these so-called intelligent algorithms must be iterated many times by probabilistic and heuristic rules, and then the optimal results are gradually obtained. Without enough iteration, undesirable or even bad results are usually yielded. That's to say, adequate iterations are required before the desired results are obtained. Otherwise, any intelligence of algorithms will not be reflected. As we all know, the time of the vertical handoff decision process should be as short as possible from the viewpoints of both end-user and network operator. Hence, it limits the application of these algorithms in the context of network selection.

Unlike the above-mentioned intelligent algorithms, the MADM algorithms are simple and straightforward and without random factors in the whole runtime. Since there are no stochastic factors, they can obtain the definite result almost directly, relying on their corresponding formulas rather than multiple loops. Although the results may be less intelligent, they can quickly complete the vertical handoff decision process and select the best target network. Therefore, these MADM algorithms still have very good advantages and provide a promising prospect in this field.

It's true that various MADM based methods have been studied in the context of network selection to rank the alternatives, but there is still a lack of a scheme based on hybrid MADM methods getting the rank list of all alternatives from the points of view of network attribute and user preference. Meanwhile, the relative importance of each relevant algorithm in calculating the final result can also be changed flexibly. Hence, this paper integrates three simple MADM algorithms (improved FAHP, Entropy and TOPSIS) and optimizes the flow of operation so as to get the comprehensive utility values of all candidate networks from the points of view of network attribute and user preference, then with the help of a threshold determines whether the mobile terminal should switch to the new network with the maximum utility value or still stay in the current one.

In summary, the major contributions of this paper are as follows.

(1) Propose an algorithm combined with FAHP, Entropy and TOPSIS to calculate the comprehensive utility value of each candidate network from the points of view of network attribute and user preference. Meanwhile, use a threshold to avoid some unnecessary vertical handoffs further.
(2) When constructing the fuzzy consistent matrix of FAHP, we make full use of its hierarchy model and adopt a novel 0.5–0.95 scale method and a formula to get the corresponding vectors.
(3) The adjustment coefficients can be flexibly adjusted according to specific traffic class or actual need. Moreover, with the change of these coefficients, the proportion of each algorithm in the final result can also be adjusted accordingly.

The rest of this paper is organized as follows. Section 2 reviews the related work on hybrid MADM methods for network selection. Section 3 describes the specific composition of the heterogeneous networks simulation scenario and the whole algorithm flow. And then the detailed process of calculating the comprehensive utility value of each network is presented. Moreover, an optimal network is selected with the help of a threshold. In Section 4, we set all parameters (network parameters and adjusting coefficients for four traffic classes) for the related simulation experiments, then test and evaluate our algorithm. Finally, conclusions and some future extensions of this proposed work are provided in Section 5.

2. Related work

For the heterogeneous network selection problem, various schemes have been proposed in the existing literature. Since the MADM methods are simple and straightforward in comparison with other non-MADM ones, many methods based on MADM are viewed as a promising direction and have been extensively studied. In this paper, we primarily focus on these MADM methods.

Chandavarkar et al. [24] propose a selection approach named SI-MAAR (Simplified and Improved Multiple Attributes Alternate Ranking) to overcome the rank reversal problem of many classical MADM methods, thereby obtaining perfect network selection reliability. On the whole, this approach itself is similar to TOPSIS. Its salient features different from TOPSIS are as follows:

(1) Use the original decision matrix and the expectation of each attribute (related to network, user, mobile device and the traffic class) to form closeness index (utility) matrix, which is equivalent to the weighted normalization decision matrix of original TOPSIS. Hence, both attribute normalization and weight calculation are removed.
(2) Each element in the positive and negative ideal solutions is 1 or 0, rather than the value in the weighted normalization decision matrix, thereby convenient to compute the Euclidean distances from each alternative network to positive or negative ideal solutions.

It can be seen that the expectation of each attribute considered in the approach for specific traffic class plays a vital important part in constructing the closeness index matrix. The performance of SI-MAAR can be affected or even degraded by the expectation of each attribute. However, opinions about the proper expectation value of each attribute obviously vary from person to person. Consequently, it means that the end-user must be very proficient with each attribute and its corresponding unit before getting a group of proper expectations for all attributes.

To our best knowledge, improvements in a single algorithm are limited after all. Meanwhile, the current heterogeneous network selection algorithms, widely used in the model of MADM, often consist of two or more simple algorithms to determine the optimal network. After giving an above example of improving single MADM method, we mainly focus on reviewing some existing hybrid MADM methods.

In [25], the authors use AHP and GRA in turn to get subjective weights of criteria and rank the networks. Similarly, a selection scheme comprising AHP and TOPSIS is proposed in [26]. The

two proposed schemes are simple and present low computational complexity of implementation, but the step of getting the relative weights of criteria completely depends on the corresponding decision maker's subjective judgments. Therefore, the final ranking of the alternatives tends to be influenced by the decision maker's experience, intuition and so on.

The authors in [27] combine AHP and bankruptcy game to calculate the potential contribution ratio (PCR) of each candidate network and select the one with the maximum PCR. Its main goal is to map the context of network selection into the bankruptcy game model: each alternative network and attribute is modeled a player and a kind of resource (or asset). Any subset of all alternatives is a coalition of players. Moreover, the normalized decision matrix is considered as resource matrix (RM). Then on the base of RM, the Shapley values can be computed for each alternative on all attributes and form a matrix denoting detailed distribution on each asset that each player can gain. At last, use the matrix above and the weight vector to compute the PCR ranking list. Obviously, for any game theory, the key step is to turn the network selection model into a corresponding processing model. The computational complexity of algorithm is much higher than that of MADM.

In [28], a vertical handover decision scheme composed of two stages is developed. SAW is used in the first stage to periodically get the required data rate (θ) for different traffic classes by weight and data rate of application. Vertical handover will not be triggered until the condition, $\theta <$ the predefined threshold, is met. Once vertical handover is triggered, the second phase of scheme will be started. This means mobile terminal would gather the information about all available networks to form a decision matrix. Then, follow the normal TOPSIS process to rank the alternatives. Both the weight for each traffic class in SAW and the weight vector of attributes in TOPSIS can be assigned directly or by default, thereby possibly affecting the performance of the scheme.

Zhou et al. [29] provide a scheme consisting of MEW and Entropy. There are the following three steps. First, use the entropy weighting method to get the objective weight of each attribute, and then calculate the combination weight with the objective weight above and the subjective weight. At last, use MEW and the combination weight to get the final ranking of alternatives. Because both the objective weight and the subjective weight are considered, the scheme outperforms SAW, TOPSIS, and GRA in terms of network load balancing and number of handovers. However, its subjective weight is assigned directly by experience alone and traffic classes are also ignored.

The authors in [17] apply AHP and VIKOR to get the weight vector of attributes for each traffic class and rank the alternatives respectively, with Network Simulator NS-3. Similarly, the authors in [30] also combine the two algorithms for network selection with NS-2. Because there are many parameters measured approximately in the context of sparse vehicular networks, the key difference from [17] is that the latter improves VIKOR so as to handle decision matrix with interval numbers. Compared with the original VIKOR, the improved VIKOR is more suitable for this issue.

In addition, the authors in [31] propose two algorithms, namely FAHP-TOPSIS and FAHP-VIKOR. After getting the weights of attributes from FAHP with the triangular fuzzy number, they use TOPSIS and VIKOR with the weights above to rank the networks respectively. The paper believes that the computational complexity of VIKOR is lower than that of TOPSIS, but with similar performance.

In [32], the authors first use the variance-based and the simplified structure inspired by a trigger-based method (named TRUST) to obtain the subjective and objective weights of attributes respectively and further get the comprehensive weights, and then employ the sigmoidal model with invariant parameter setting to adjust the utilities of the normalized attributes, finally utilize SAW to obtain the final comprehensive ranking. However one of the disadvantages of this method is that the proportion of objective and subject weight cannot be adjusted dynamically for some requirements.

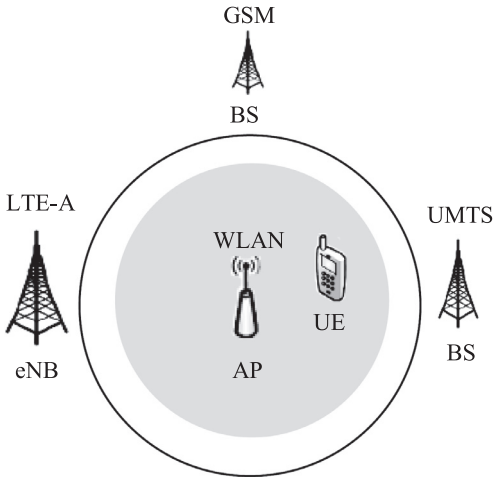


Fig. 1. Simulation scenario of network selection.

In [33], combining FAHP, Entropy and GRA, the authors propose a new scheme named FAHPE-GRA, which utilizes the first two algorithms to calculate the subjective and objective weights respectively and then employs least squares method to get the comprehensive weights of attributes, finally ranks the networks by GRA. In FAHP, construct fuzzy comparison matrix with triangular fuzzy numbers (TFNs) according to the way creating an AHP pair-wise comparison matrix, then use a given equation to transform each TFNs in fuzzy comparison matrix into a crisp number. Next, follow the AHP algorithm completely. Such the FAHP algorithm is suitable to solve the MADM problem with non-numerical attributes, but still has the shortcoming of AHP. If all the considered attributes are numerical, then there is no need to apply the FAHP algorithm similar to AHP. At this time, to overcome the disadvantages of AHP, we can use another FAHP algorithm, which constructs fuzzy consistent matrix rather than pair-wise comparison matrix. Hereafter the FAHP algorithm refers to such algorithm.

Compared with the aforementioned works in this area, the proposed optimal scheme integrating FAHP, Entropy and TOPSIS, makes full use of the hierarchy model of FAHP to determine subjective weight vector of attributes and subjective utility values of user preferences, and uses the Entropy method to get objective weight vector of attributes, and uses TOPSIS to get objective utility values of user preferences. In addition, by groups of adjustment coefficients with respect to specific traffic classes, the final rank list of all alternatives is obtained. At last, use a threshold to let mobile terminal stay in its current network as long as possible, rather than blindly handover to a new network which is just a little better than the current network, thereby avoiding the ping pong effect further.

3. Algorithm model formulation

As shown in Fig. 1, in a heterogeneous network scenario comprised of four disparate candidate networks (i.e., GSM, UMTS, LTE-A, WLAN), the mobile terminal (named UE) is assumed to move slowly in random motion within the WLAN range. Obviously for the four networks above, only WLAN is a local area wireless communication network, and the rest are wide area wireless communication networks. Consequently, we suppose that the signals of GSM, UMTS and LTE-A completely cover the whole simulation area,

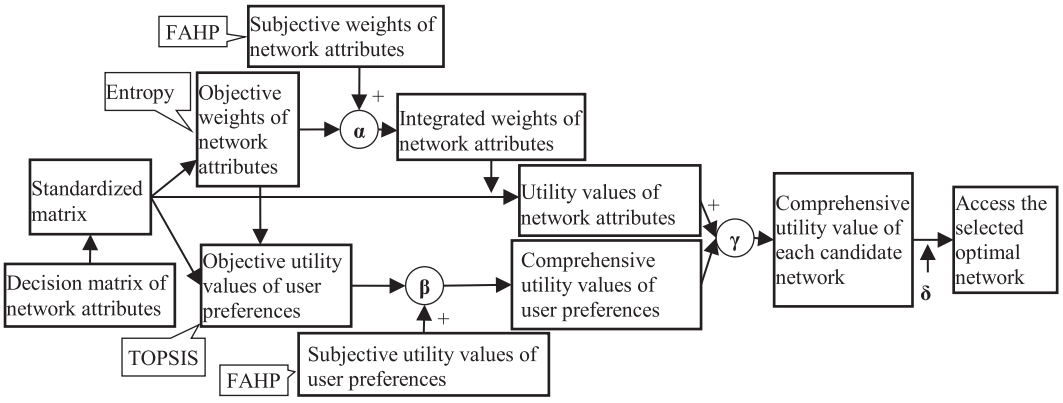
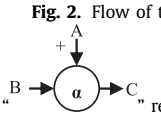


Fig. 2. Flow of the proposed algorithm.



Note: In Fig. 2, such a symbol " " represents the arithmetic operation: $C = \alpha \times A + (1-\alpha) \times B$.

while the larger circular and the smaller gray circular areas are the coverage of WLAN signal and the moving area of the UE respectively. The UE can run four typical traffic classes, including conversational, streaming, interactive and background. For the sake of simplicity, the conversational class herein refers to the voice session class by default.

To facilitate simulation experiments with the MADM algorithms, we assume that the UE contains a function module, which can run periodically to detect and collect all the necessary information about six parameters of the heterogeneous networks surrounding the UE. The trade-off parameters of each alternative network include available bandwidth (B), latency (D), delay jitter (J), packet loss ratio (L), bit error rate (E) and service price (C).

Thus, in a heterogeneous network scenario shown in Fig. 1, the network selection process can be formally described as follows: given the set of candidate networks *Nets* = {GSM, UMTS, LTE-A, WLAN}, the set of decision indicators (namely network attributes) *Attributes* = {B, D, J, L, E, C}, the set of traffic classes *TrafficClasses* = {conversational class, interactive class, streaming class, background class}, corresponding adjustment coefficients α, β, and γ for each traffic class, and the threshold δ which is used to decide whether the UE should switch to the network with the largest comprehensive utility value. As shown in Fig. 2, the whole process consists of the following four parts.

Firstly, before simulating, we use FAHP to calculate both network attributes' subjective weights and user preferences' subjective utility values related to each traffic class. The results and corresponding adjustment coefficients are stored in the UE. In the simulation process, the UE can detect the running traffic class and network parameters. According to the traffic class, the UE easily gets the corresponding information (i.e.: network attributes' subjective weights; user preferences' subjective utility values; and the three adjustment coefficients α, β, and γ) from its memory to prepare for calculating the comprehensive network utility value of each candidate network. Meanwhile, a decision matrix with the six parameters of every candidate network can be formed and standardized.

Next, the objective utility value of user preference is obtained by combining the TOPSIS approach and the network attributes' objective weights, which are calculated by the Entropy method.

Then, we use α to integrate the network attributes' subjective and objective weights to get the integrated weights of network attributes, and use β to integrate the user preferences' subjective and objective utility values to get the comprehensive utility values of user preferences. Furthermore, the comprehensive utility value of each candidate network is calculated in terms of network attribute and user preference with γ.

Finally, according to the result whether the ratio of the maximum utility value and the utility value of the current network is larger than the threshold δ, we make a decision that the UE should stay in the current network or switch to the new network with the maximum utility value.

As shown in Fig. 2, with the change of these adjustment coefficients (α, β, and $\gamma \in [0, 1]$), the proportion of each algorithm in the final result can also be adjusted accordingly. In this regard, we discuss the following.

(1) If $\alpha = 0$ or $\beta = 0$, then the integrated weights of network attributes and the comprehensive utility values of user preferences will be only made of the objective weights of network attributes and the objective utility values of user preferences respectively. Moreover, if α and β are zero simultaneously, FAHP will not work. With the decrease of α and β and, the proportion of FAHP will decrease, and vice versa. Similarly, if $\alpha = 1$ or $\beta = 1$, Entropy or TOPSIS will not work.
(2) If $\gamma = 0$ or $\gamma = 1$, then the final rank list of all alternatives is determined by the angle of user preference or network attribute respectively.
(3) If any of the adjustment coefficients is 0.5, then the related two parts are equally important.

According to the discussion above, none of the adjustment coefficients is 0 or 1, which means that the three algorithms work together to compute the comprehensive utility value of each candidate network from the two viewpoints of network attribute and user preference. Thus, to evaluate the performance consisting of the three algorithms simultaneously, we set the adjustment coefficients (α, β, and γ) are larger than 0 and smaller than 1.

In addition, without loss of generality, we assume the number of alternative networks, attributes and traffic classes are *m, n, t*. Clearly in our heterogeneous network scenario, their values are 4, 6 and 4, respectively.

According to the characteristics of each element of *TrafficClasses* and the user preferences, we use FAHP to calculate the subjective weights of network attributes and the subjective utility values of user preferences, and then saved them in the UE. Once the traffic class running in the UE is detected, both the relevant subjective information (weights of network attributes and utility values of user preferences) and the corresponding adjusting parameters (α, β, and γ) can be obtained. After the standardization process of each decision matrix formed by the UE, the objective weights of

Table 1
0.5–0.95 measure for elements of fuzzy consistent matrix.

Compare the importance of x_i and x_j	Equally important	Moderately important	Obviously important	Strongly important	Especially important	Extremely important
a_{ij}	0.5	0.6	0.7	0.8	0.9	0.95

Note: The four values (0.55, 0.65, 0.75 and 0.85) indicate the intermediate values of the adjacent judgments above; $a_{ij} < 0.5$ illustrates x_i is less important than x_j; and while the comparison of x_j and x_i, namely a_{ji}, is 1-a_{ij}.

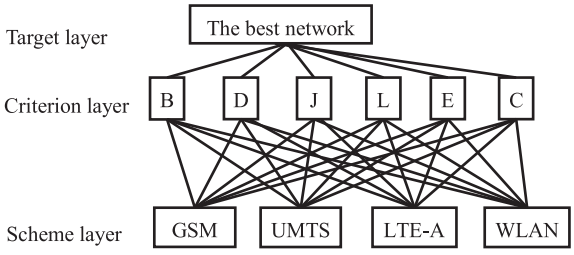


Fig. 3. Hierarchy model of FAHP.

network attributes and the objective utility values of user preferences are calculated through the Entropy method and TOPSIS respectively. Then, combining α, β, and γ, the comprehensive utility value of each candidate network can be determined. Finally, according to the threshold δ, we can determine the UE should either access the network with the maximum comprehensive utility value or continue to stay in the current network.

3.1. Use FAHP to calculate subjective attribute weights and subjective utility values of user preferences

At first, we establish the typical hierarchical structure model for FAHP, as shown in Fig. 3.

In order to enhance the description ability and the discrimination of the FAHP algorithm to the minimum and maximum attribute weights in the heterogeneous network selection problem, we improve the standard FAHP algorithm in the following two aspects:

(1) When comparing the importance of x_i and x_j (i.e. the element of fuzzy consistent matrix a_{ij}) ranging from "equally important" to "extremely important", we use the famous thought of 1–9 scale method for reference, which is used to set the elements of pairwise comparison matrix in AHP, and adopt a novel 0.5–0.95 scale method, as shown in Table 1.
(2) The elements of the first column of the fuzzy consistent matrix reflect the relative importance of each attribute to a certain extent. Thus we use Eq. (1) to calculate the weight of the element x_i, basing on the original formula provided by [34].

$$w_i = \frac{\sum_{j=1}^{n} a_{ij} + \frac{n}{2} - 1}{n(n-1)} \times (2a_{i1})^k, k \geq 1 \quad (1)$$

where n is the number of attributes and k is a constant parameter. In this paper, $k = 2$.

According to the different requirements of each traffic class in *TrafficClasses* for each attribute in *Attributes* in the relationship between the Target layer and the Criterion layer, we set up four fuzzy consistent matrices for four traffic classes in *TrafficClasses* (namely conversational, streaming, interactive and background). The four matrices are denoted as *Ac, As, Ai* and *Ab* respectively. And then the corresponding weight vectors (i.e., *WAc, WAs, WAi* and *WAb*) can be directly obtained by Eq. (1), as shown in Tables 2–5. Meanwhile, according to the pairwise comparison of the importance of networks in *Nets* for each attribute in *Attributes* in the relationship between the Criterion layer and the Scheme Layer, we establish six fuzzy consistent matrices for these attributes (namely available bandwidth, latency, delay jitter, packet loss ratio, bit error rate and service price). The six matrices are denoted by *Fb, Fd, Fj, Fl, Fe* and *Fc* respectively. And then the corresponding weight vectors (i.e., *WPb, WPd, WPj, WPl, WPe* and *WPc*) are obtained by Eq. (1) and illustrated in Tables 6–11.

Table 2
Fuzzy consistent matrix and weight vector for conversational class.

Ac	B	D	J	L	E	C	WAc
B	0.5	0.1	0.2	0.4	0.4	0.5	0.0708
D	0.9	0.5	0.6	0.8	0.8	0.9	0.3637
J	0.8	0.4	0.5	0.7	0.7	0.8	0.2609
L	0.6	0.2	0.3	0.5	0.5	0.6	0.1169
E	0.6	0.2	0.3	0.5	0.5	0.6	0.1169
C	0.5	0.1	0.2	0.4	0.4	0.5	0.0708

Table 3
Fuzzy consistent matrix and weight vector for streaming class.

As	B	D	J	L	E	C	WAs
B	0.5	0.3	0.6	0.75	0.6	0.7	0.2003
D	0.7	0.5	0.8	0.95	0.8	0.9	0.4790
J	0.4	0.2	0.5	0.65	0.5	0.6	0.1141
L	0.25	0.05	0.35	0.5	0.35	0.45	0.0363
E	0.4	0.2	0.5	0.65	0.5	0.6	0.1141
C	0.3	0.1	0.4	0.55	0.4	0.5	0.0562

Table 4
Fuzzy consistent matrix and weight vector for interactive class.

Ai	B	D	J	L	E	C	WAi
B	0.5	0.75	0.85	0.4	0.4	0.5	0.1855
D	0.25	0.5	0.6	0.15	0.15	0.25	0.0335
J	0.15	0.4	0.5	0.05	0.05	0.15	0.0102
L	0.6	0.85	0.95	0.5	0.50	0.6	0.2968
E	0.6	0.85	0.95	0.5	0.50	0.6	0.2370
C	0.5	0.75	0.85	0.4	0.4	0.5	0.2370

Table 5
Fuzzy consistent matrix and weight vector for background class.

Ab	B	D	J	L	E	C	WAb
B	0.5	0.45	0.5	0.1	0.05	0.5	0.0648
D	0.55	0.5	0.55	0.15	0.1	0.55	0.0842
J	0.5	0.45	0.5	0.1	0.05	0.5	0.0648
L	0.9	0.85	0.9	0.5	0.45	0.9	0.3332
E	0.95	0.9	0.95	0.55	0.5	0.95	0.3882
C	0.5	0.45	0.5	0.1	0.05	0.5	0.0648

Table 6
Fuzzy consistent matrix and weight vector for B.

B	GSM	UTMS	LTE-A	WLAN	WPb
GSM	0.5	0.3	0.15	0.1	0.0696
UTMS	0.7	0.5	0.35	0.3	0.1898
LTE-A	0.85	0.65	0.5	0.45	0.3388
WLAN	0.9	0.7	0.55	0.5	0.4018

Table 7
Fuzzy consistent matrix and weight vector for D.

D	GSM	UTMS	LTE-A	WLAN	WPd
GSM	0.5	0.6	0.7	0.85	0.5245
UTMS	0.4	0.5	0.6	0.75	0.2989
LTE-A	0.3	0.4	0.5	0.65	0.1475
WLAN	0.15	0.25	0.35	0.5	0.0291

Table 8
Fuzzy consistent matrix and weight vector for J.

J	GSM	UTMS	LTE-A	WLAN	WPj
GSM	0.5	0.4	0.4	0.8	0.2299
UTMS	0.6	0.5	0.5	0.9	0.3738
LTE-A	0.6	0.5	0.5	0.9	0.3738
WLAN	0.2	0.1	0.1	0.5	0.0225

Table 9
Fuzzy consistent matrix and weight vector for L.

L	GSM	UTMS	LTE-A	WLAN	WPl
GSM	0.5	0.6	0.7	0.8	0.5114
UTMS	0.4	0.5	0.6	0.7	0.2909
LTE-A	0.3	0.4	0.5	0.6	0.1432
WLAN	0.2	0.3	0.4	0.5	0.0545

Table 10
Fuzzy consistent matrix and weight vector for E.

E	GSM	UTMS	LTE-A	WLAN	WPe
GSM	0.5	0.6	0.7	0.8	0.5114
UTMS	0.4	0.5	0.6	0.7	0.2909
LTE-A	0.3	0.4	0.5	0.6	0.1432
WLAN	0.2	0.3	0.4	0.5	0.0545

Table 11
Fuzzy consistent matrix and weight vector for C.

C	GSM	UTMS	LTE-A	WLAN	WPc
GSM	0.5	0.55	0.7	0.25	0.1992
UTMS	0.45	0.5	0.65	0.2	0.1506
LTE-A	0.3	0.35	0.5	0.05	0.0526
WLAN	0.75	0.8	0.95	0.5	0.5976

Table 12
Subjective utility values of user preferences.

UPS	GSM	UTMS	LTE-A	WLAN
Conversational class	0.3894	0.2983	0.2123	0.1000
Streaming class	0.3795	0.2761	0.2056	0.1388
Interactive class	0.3530	0.2400	0.1605	0.2465
Background class	0.4453	0.2813	0.1653	0.1081

In order to facilitate subsequent calculations, we put the column vectors of subjective weights for t traffic classes into a matrix $WAS=(\mathrm{was}_{ij})_{n\times t}=[WAc\ WAs\ WAi\ WAb]$ and the n column vectors of subjective weights of user preferences into a matrix $WPS=(\mathrm{wps}_{ij})_{m\times n}=[WPb\ WPd\ WPj\ WPl\ WPe\ WPc]$.

Then the subjective utility values of user preferences, $UPS=(ups_{ij})_{t\times m}$, are calculated by Eq. (2):

$$UPS=(WPS\times WAS)^{T} \tag{2}$$

Then we can easily get a two-dimensional matrix *UPS* tabulated in Table 12.

3.2. Standardize the decision matrix

The decision matrix of network parameters detected by the UE is denoted as $R=(r_{ij})_{m\times n}$. The elements of *Attributes* can be divided into two categories: benefit type and cost type, which are denoted as *AT1* and *AT2* respectively. Obviously, all attributes belong to *AT2* except the available bandwidth. To remove incommensurability and contradictions between different types of attributes, it is necessary to take two steps to normalize or standardize the original *R*.

First, use Eq. (3) to process *R* to convert all non-benefit attributes into ones belonging to *AT1*. For convenience, the new matrix is still recorded as *R*.

$$r_{ij}=\begin{cases}\dfrac{r_{ij}}{\min\limits_{i}(r_{ij})+\max\limits_{i}(r_{ij})}, & x_j\in AT1\\ 1-\dfrac{r_{ij}}{\min\limits_{i}(r_{ij})+\max\limits_{i}(r_{ij})}, & x_j\in AT2\end{cases},\ 1\le i\le m, 1\le j\le n \tag{3}$$

where $\min\limits_{i}(r_{ij})$ and $\max\limits_{i}(r_{ij})$ respectively represent the minimum and maximum value of the *j*th column elements of *R*.

Second, according to Eq. (4) the new matrix (still referred to as *R*) is obtained by the normalized processing of each column of *R*.

$$r_{ij}=\frac{r_{ij}}{\sum_{i=1}^{m} r_{ij}} \tag{4}$$

3.3. Use entropy to compute the objective weights of network attributes

The "entropy" is a widely used concept in thermodynamics, e.g. use it to measure the instability of a system. After introduced to the field of information theory, entropy value indicates the degree of difference in the system. The greater the difference among the specific values on an attribute of all candidate networks, the smaller the entropy value of the attribute. The greater the amount of information contained in it, the greater the weight occupied by it, and vice versa. Thus, the Entropy method can be used to get the weights of attributes. Because there is no human interference, it is an objective weighting method. The detailed process of calculating the objective weights of attributes with the Entropy method is as follows:

First, use the obtained *R* by Eq. (4) to calculate the entropy value of each attribute by Eq. (5).

$$E_j=-k\sum_{i=1}^{m} r_{ij}\ln(r_{ij}),\ 1\le j\le n \tag{5}$$

where: E_j represents the entropy value of the *j*th attribute; and *k* is a constant parameter, and herein it takes $\frac{1}{\ln(m)}$.

Second, obtain the objective weights of attributes using Eq. (6).

$$wao_j=\frac{1-E_j}{n-\sum_{j=1}^{n} E_j},\ 1\le j\le n \tag{6}$$

where wao_j represents the objective weight of the *j*th attribute.

Furthermore, the objective weights of *n* attributes form a column vector $WAO=[wao_1\ \mathrm{wao}_2\ \ldots\ wao_n]^{\mathrm{T}}$. Obviously, *WAO* is independent of the traffic class running in the UE and only related to the network parameters decision matrix detected by the UE.

3.4. Obtain the objective utility values of user preferences with TOPSIS

TOPSIS, widely used at present, is also a kind of objective algorithm in MADM. It often uses geometric distance space (e.g. Euclidean distance) as the distance between the two candidates, and the best option is the one which is both the nearest away from the

ideal scheme and farthest away from the worst scheme [35]. The ideal scheme and the worst scheme, also known as positive ideal solution and negative ideal solution respectively, consist of the best value and the worst value of each attribute separately. Clearly, both the positive and the negative ideal solution are fictional in order to facilitate the calculation of the real extent of the merits of all candidates [36,37]. We use TOPSIS combined with the objective weights of attributes above obtained by the Entropy method to determine the objective utility values of user preferences. The four steps are as follows:

First, the positive and negative ideal solutions, $R^+ = (r_j^+)_{n\times 1}$ and $R^- = (r_j^-)_{n\times 1}$, are obtained by Eq. (7) and R which has been standardized by Eq. (4).

$$\begin{cases} r_j^+ = \max\limits_{1\le i\le m}(r_{ij}) \\ r_j^- = \min\limits_{1\le i\le m}(r_{ij}) \end{cases}, 1 \le j \le n \tag{7}$$

Because all element values of R have been unitless and normalized into the interval [0, 1], there is no need to use Eq. (8) below to normalize the original decision matrix again, in contrast with the original TOPSIS method [37]. As a result, we normalize the decision matrix uniformly once for the Entropy method and TOPSIS, and thus reduce computation cost.

$$r_{ij} = \frac{r_{ij}}{\sqrt{\sum_{i=1}^{m} r_{ij}^2}} \tag{8}$$

Second, the weighted Euclidean distances of each alternative network to R^+ and R^-, respectively denoted as $D^+ = (d_i^+)_{m\times 1}$ and $D^- = (d_i^-)_{m\times 1}$, are given as:

$$\begin{cases} d_i^+ = \sqrt{\sum\limits_{j=1}^{n} wao_j (r_{ij} - r_j^+)^2} \\ d_i^- = \sqrt{\sum\limits_{j=1}^{n} wao_j (r_{ij} - r_j^-)^2} \end{cases}, 1 \le i \le m \tag{9}$$

where wao_j represents the objective weight of the jth attribute obtained by Eq. (6).

Third, use Eq. (10) to calculate the relative similarity (denoted as $S = (s_i)_{m\times 1}$) between each network and the virtual optimal one represented by the positive ideal solution.

$$S_i = \frac{1}{1 + \frac{d_i^+}{d_i^-}}, 1 \le i \le m \tag{10}$$

Fourth, normalize S to calculate the objective utility values $UPO = (upo_i)_{m\times 1}$ using Eq. (11).

$$upo_i = \frac{S_i}{\sum_{k=1}^{m} S_i}, 1 \le i \le m \tag{11}$$

3.5. Get the comprehensive utility values of all alternative networks

Assuming that the identifier of traffic class currently running in the UE is k ($1 \le k \le t$), it is easy to know the following information relevant to the traffic class k: the subjective weights of attributes (denoted as WAS_k) is the kth column of the matrix WAS, the subjective utility values of user preferences (denoted as UPS_k) is the kth row of the matrix UPS, and three adjustment coefficients (α, β, and γ). The four steps of calculating the comprehensive utility values of alternatives are as follows:

First, use Eq. (12) to get the integrated weights of attributes $WA = (wa_i)_{n\times 1}$.

$$WA = \alpha \times WAS_k + (1 - \alpha) \times WAO \tag{12}$$

where WAO represents the objective weights of attributes obtained by Eq. (6).

Second, use Eq. (13) to get the integrated weights of attributes $UA = (ua_i)_{m\times 1}$.

$$UA = R \times WA \tag{13}$$

where R is the standardized decision matrix by Eq. (4).

Third, use Eq. (14) to get the utility values of user preferences $UP = (up_i)_{m\times 1}$.

$$UP = \beta \times UPS_k + (1 - \beta) \times UPO \tag{14}$$

where UPO represents the objective utility values of user preferences obtained by Eq. (11).

Finally, use Eq. (15) to calculate the comprehensive utility values of networks $U = (u_i)_{m\times 1}$.

$$U = \gamma \times UA + (1 - \gamma) \times UP \tag{15}$$

3.6. Determine the optimal network to access

The descending order of comprehensive utility values of all candidate networks above can reflect the degree of the merits of networks on the whole. In other words, the network with the maximum utility value is the best. However, we believe that the strategy of always switching to the one with the largest utility value is simple but has many drawbacks, due to the fact that it will undoubtedly bring cost when vertical handover occurs in a heterogeneous network environment. It's well known that the ping-pong effect arises from switching too frequently. Furthermore, in the case that the current network of the UE meets the basic needs of the running traffic class, it is clearly that the number of vertical handovers should be controlled strictly or minimized.

If the utility value of network i, U_i, is maximum, and the utility value of the other network j ($i \ne j$ and the network j is currently attached by the UE) is U_j, then use $\triangle$ to represents the difference between U_i and U_j, i.e. $\triangle = U_i - U_j$. If the UE switches to network i under the circumstance that the value of $\triangle$ is very small, then the benefit of the net obtained from this vertical handover is likely to offset and appears "more harm than good". We define such a vertical handoff herein as "unnecessary vertical handoff". At this time, the UE had better continue adhering to the current network j instead of triggering vertical handoff. Based on the analysis above, when determining the best network the UE accesses next, we introduce a tiny threshold δ ($\delta \ge 1$) to reduce the number of unnecessary handovers. The relevant pseudo-code is described in Algorithm 1.

Algorithm 1
Get the ID of the best target network for the UE.

```
1: int getBestNID (currentNID, U, Th){
2:     // currentNID, U and Th are the ID of current network of UE, the column
       vector of comprehensive utility values of all candidate networks and the
       threshold δ determining the optimal one, respectively.
3:     int bestNID = currentNID;
4:     [maxUV, maxUID] = max(U); // maxUV and maxUID are the maximum
       value of U and its ID, respectively.
5:     if (currentNID == 0 || currentNID == maxUID || maxUV /
       U(currentNID) > Th){
6:         bestNID = maxUID;
7:     }
8:     return bestNID; // Output: the ID of the best target network
9: }
```

Note: When the UE is not connected to any network initially, the ID of its current access network simply means zero. While for the other time, its value always varies from 1 to m, as long as the UE is attaching a network.

According to the mechanism of our algorithm, we can analyze its complexity. Since it combines FAHP, Entropy, and TOPSIS, its

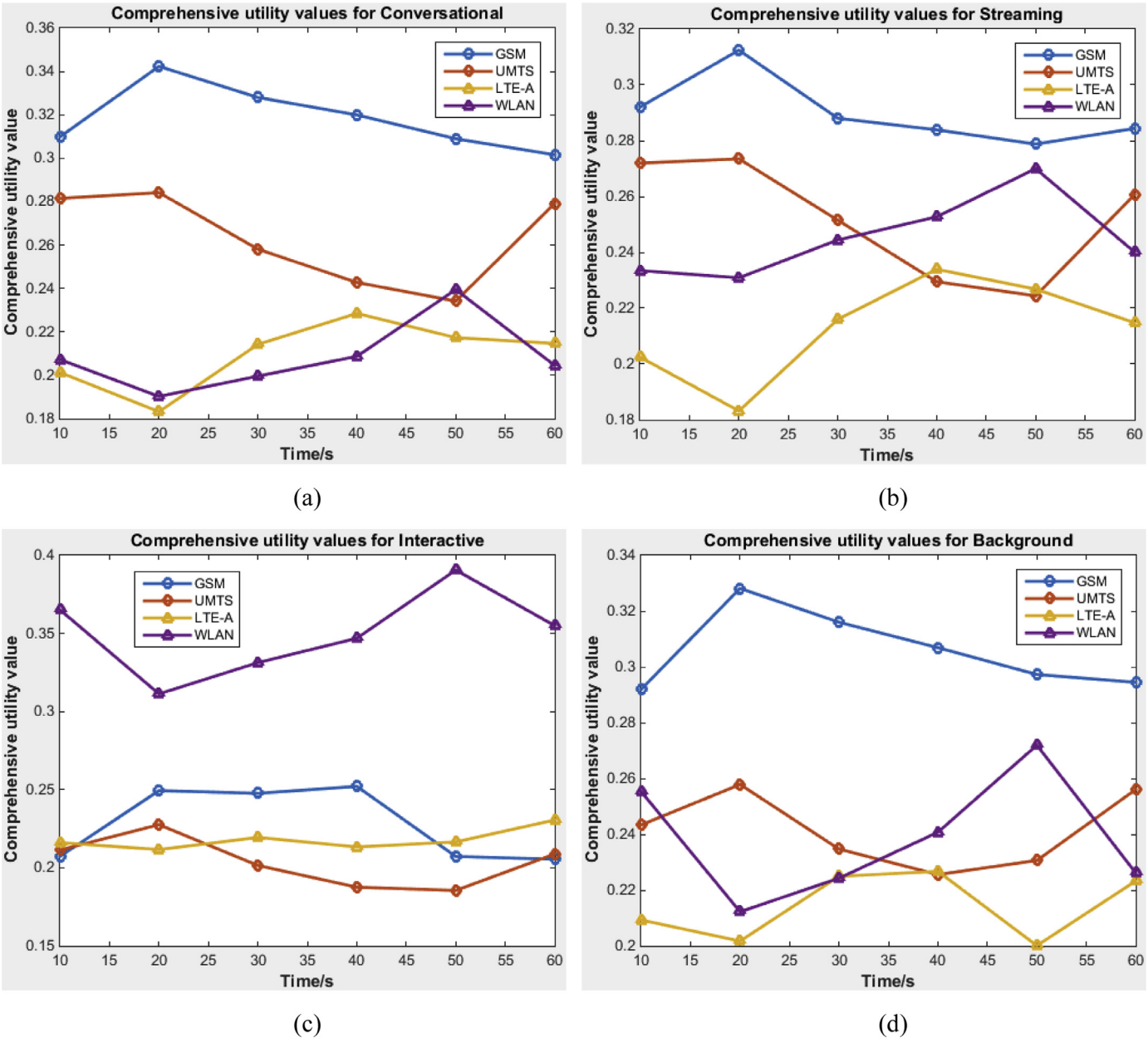


Fig. 4. Six continuous changes of comprehensive utility values of networks for the four traffic classes.

time and space complexity are much more complicated than those of the three algorithms. However, it should be noted that once the subjective weights of network attributes and utility values of user preferences are obtained by FAHP, the rest work can be done by two other algorithms. Hence, the time and space complexity are mainly determined by Entropy, and TOPSIS. Its time complexity is $O(N^3)$. As for its space complexity, it only needs to store some vectors, such as weight vectors of attributes and user preferences, the rank of networks and three adjustment coefficients for each traffic class, and so on. Therefore, from this point of view, its space complexity is $O(N^2)$. Because the values of m, n and t are very small (generally not more than 10), the complexity of the algorithm is still acceptable.

4. Performance simulation and analysis

In this section, we will test and evaluate our proposed scheme through extensive simulation experiments in two aspects. On the one hand, we find out the best network among the alternatives at a specific time for each traffic class to verify the effectiveness and correctness of the proposed algorithm. On the other hand, compare the proposed method with the three existing hybrid MADM methods, by the repeated network selection experiments in the dynamic simulation environment.

4.1. Set the simulation parameters

We use Matlab as the simulation platform. According to the scenario in Fig. 1, the values of network simulation parameters and the adjustment coefficients for the four traffic classes are shown in Tables 13 and 14 respectively.

In order to simulate the situation that the decision matrix detected by the UE has some changes when it is moving, we view the available bandwidth and the service price as static attributes, while the remaining four attributes are dynamic(namely changeable) and uniformly distributed. For instance, 45 (40–80) in the “latency" col-

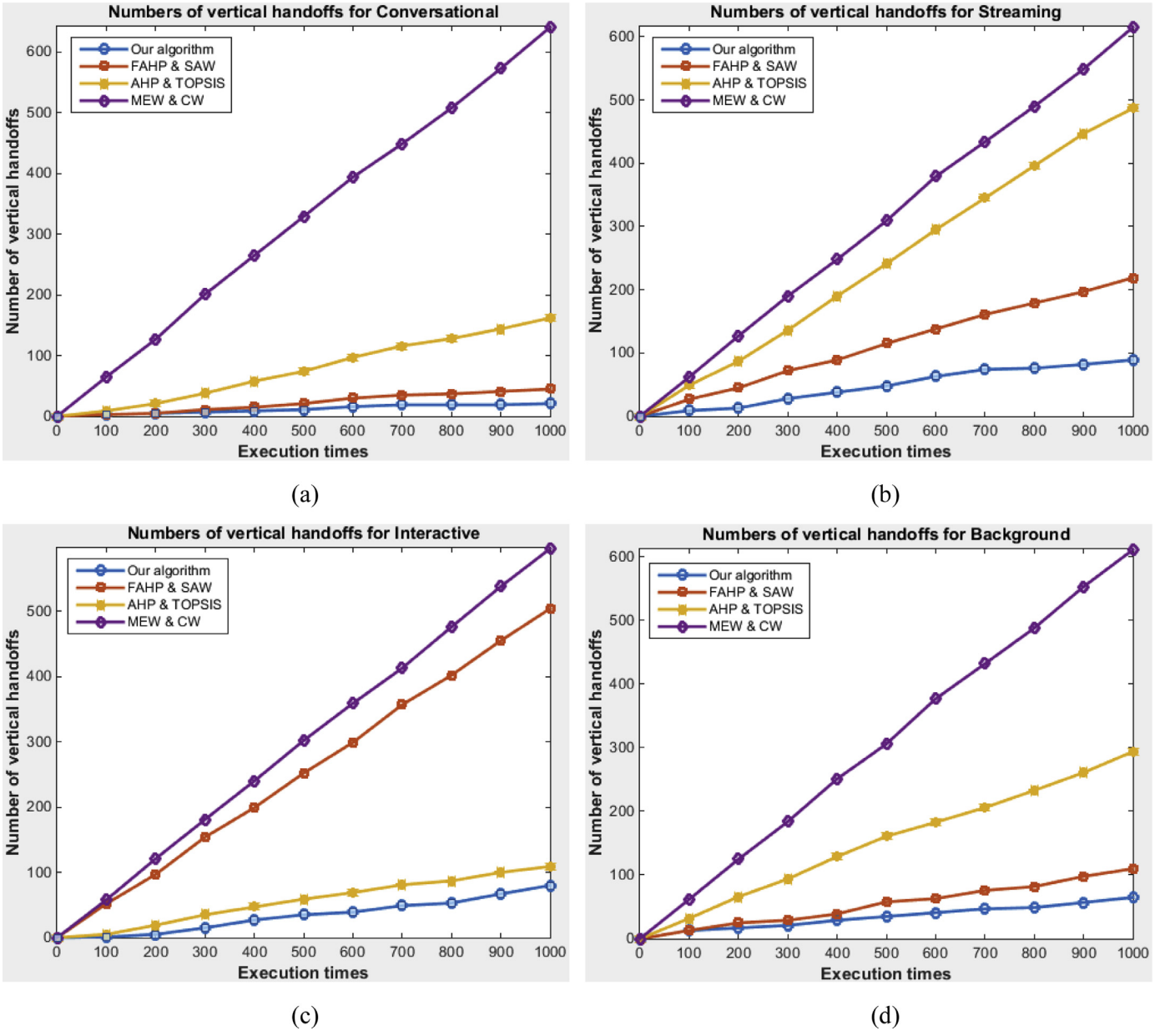


Fig. 5. Comparison of numbers of vertical handoffs of three algorithms.

Table 13
Network parameters.

Networks	B/Mbps	D/ms	J/ms	L/%	E/%	C
GSM	0.3	30(10∼60)	15(10∼40)	0.5(0.3∼1.4)	0.03(0.01∼0.05)	0.4
UMTS	1.1	45(40∼80)	30(25∼60)	0.9(0.7∼2.2)	0.05(0.04∼0.09)	0.3
LTE-A	2.0	60(70∼110)	50(45∼80)	1.2(0.8∼2.3)	0.10(0.08∼0.12)	0.2
WLAN	5.0	85(90∼140)	65(60∼100)	1.6(0.9∼2.4)	0.20(0. 10∼0.35)	0.1

Table 14
Adjustment coefficients for the four traffic classes.

Traffic class	α	β	γ
Conversational class	0.8	0.7	0.8
Streaming class	0.8	0.8	0.8
Interactive class	0.5	0.1	0.2
Background class	0.8	0.4	0.5

umn of the UMTS row in Table 13 means: the variable value of UMTS' latency attribute ξ meets: $\xi \backsim$ U [40, 80], while 45 is its snapshot value detected by the UE at a specific time(hereafter denoted as T). The rest in Table 13 can be done in the same manner.

4.2. Network selection at the single time t

According to the algorithm flow shown in Fig. 2, the comprehensive utility values of all alternatives for four traffic classes at time T are calculated, as shown in Table 15.

Table 15
Comprehensive utility values of networks.

U	GSM	UTMS	LTE-A	WLAN
Conversational class	**0.3146**	0.2669	0.2256	0.1929
Streaming class	**0.2853**	0.2532	0.2283	0.2332
Interactive class	0.2189	0.2099	0.2267	**0.3445**
Background class	**0.3049**	0.2541	0.2184	0.2226

We assume the UE has not yet accessed to any network at time T. From Table 15, for the four traffic classes the selected networks are GSM, GSM, WLAN and GSM respectively. According to the requirements of the attributes in *Attributes* for the four traffic classes, at time T, the latency and delay jitter of GSM are the smallest, so it is suitable for the real time traffic classes, namely conversational and streaming class. On the other hand, although the bandwidth of three other networks is higher than that of GSM, their latency and delay jitter are significantly higher than GSM. Consequently, they are not suitable for real-time traffic and our algorithm selects GSM to run the streaming class. With the largest bandwidth and the lowest price, WLAN is very suitable for running the interactive class. But both the bit error rate and packet loss ratio of GSM are the lowest, so it is suitable for running the background class. In summary, at time T, our algorithm can make appropriate selection results for four traffic classes, reflecting its rationality and correctness.

4.3. Network selections at many times

Assuming that the network parameters detecting module in UE repeats every 10 s, the comprehensive utility values of the candidate networks are changed as shown in Fig. 4, where (a)–(d) respectively represents six changes within 60 s for the four traffic classes in the UE, namely conversational, streaming, interactive, and background class.

It can be seen from Fig. 4 that in the simulation period of 60 s, the networks with the largest comprehensive utility values for the four traffic classes are GSM, GSM, WLAN and GSM respectively. It not only corresponds with the conclusion above at time T, but also shows that at the six time points the network with the largest utility value remains significantly higher than the rest. The results show that, with the help of the comprehensive utility values and the parameters given in Table 14, the final selection result shows good stability and reliability.

4.4. Performance comparison

For the purpose of illustrating the superiority of the proposed algorithm, we use other three hybrid algorithms (FAHP & SAW, AHP & TOPSIS [26] and MEW & CW [29], hereinafter referred to as Algorithms 1, 2 and 3, respectively) for comparison. To be fair, the subjective weights of network attributes of the four algorithms for each of the four traffic classes are the same. Let the threshold δ take 1.02, and the numbers of vertical handoffs of the four algorithms in 1000 network selections for the four traffic classes are respectively shown in (a) to (d) of Fig. 5.

From Fig. 5, it can be seen that via 1000 network selections for each traffic class, the numbers of vertical handoffs of the four algorithms all increase with the times of network selections at different speeds. In this respect, the Algorithm 3 and our algorithm are the fastest and the slowest for the four traffic classes, respectively. And except the interactive class, both the Algorithm 1 and our algorithm are superior to the other two algorithms. But only our algorithm alone can grow in a relatively gentle way rather than increase sharply. It shows that our algorithm can obtain the relatively stable comprehensive utility values of networks from two

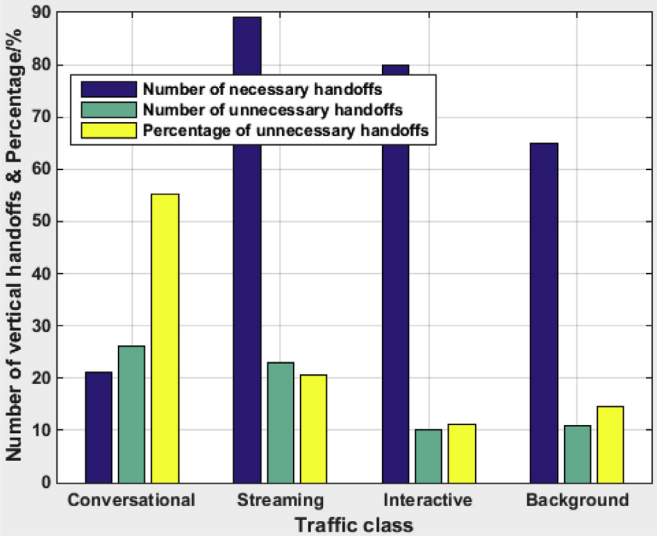


Fig. 6. Comparison of number of necessary & unnecessary handovers for four traffic classes.

aspects of network attribute and user preference, thereby avoiding the UE switching frequently between networks due to some slight changes of network parameters, which cause large changes in the final ranking. Therefore, our algorithm has some remarkable advantages in reducing the total number of vertical handoffs. In addition, the comprehensive utility values implying the subjective and objective factors and the reasonable threshold δ can decrease some unnecessary vertical handovers (as shown in Fig. 6). Clearly, the bigger the threshold δ is, the more obvious the effect of filtering out unnecessary handovers is.

From Fig. 6, it can be seen that the percentage of unnecessary handovers for conversational class is the highest, with a value of more than 50%. While all the corresponding values of the other three classes are lower than 21%. This also proves the fact that for conversational class, the UE is particularly apt to stay in the current network as long as possible, compared with three other traffic classes. By contrast, for interactive and background classes, the UE is eager to switch to a better network, ignoring the potential risk of ping-pong effect. This interesting phenomenon also relevantly reflects people's general view of the four traffic classes.

Apart from the terms of number of handovers above, the quantified benefit from vertical handoff (named "gains") is defined by Eq. (16) to compare the performance of the three algorithms further.

$$G_{ij}^{s} = \lambda_{ij}^{s} \prod_{k=1}^{n} r_{jk} w_k^s,\ 1 \le i, j \le m,\ 1 \le s \le t \tag{16}$$

where s is the ID of traffic class running in the UE, n is the number of attributes, G_{ij}^{s} represents the gains from the vertical handover from current network i to target network j, for the traffic class s. And similarly, λ_{ij}^{s} represents the corresponding negative influence factor related to the handover above, $r_{jk}$is the element of R by Eq. (4), and w_k^s is the subjective weight of attribute k for the traffic class s. To be fair, $\lambda_{ij}^{s}$is given by Eq. (17).

$$\lambda_{ij}^{s} = \begin{cases} 1, & i = j \\ const, & i \ne j \end{cases} \tag{17}$$

Let λ_{ij}^{s} take 0.8, then the average values of gains of three algorithms in 1000 network selections for the four traffic classes are respectively shown in (a)–(d) of Fig. 7.

From Fig. 7, it can be seen that the average values of gains of our algorithm have an advantage over those of three other algo-

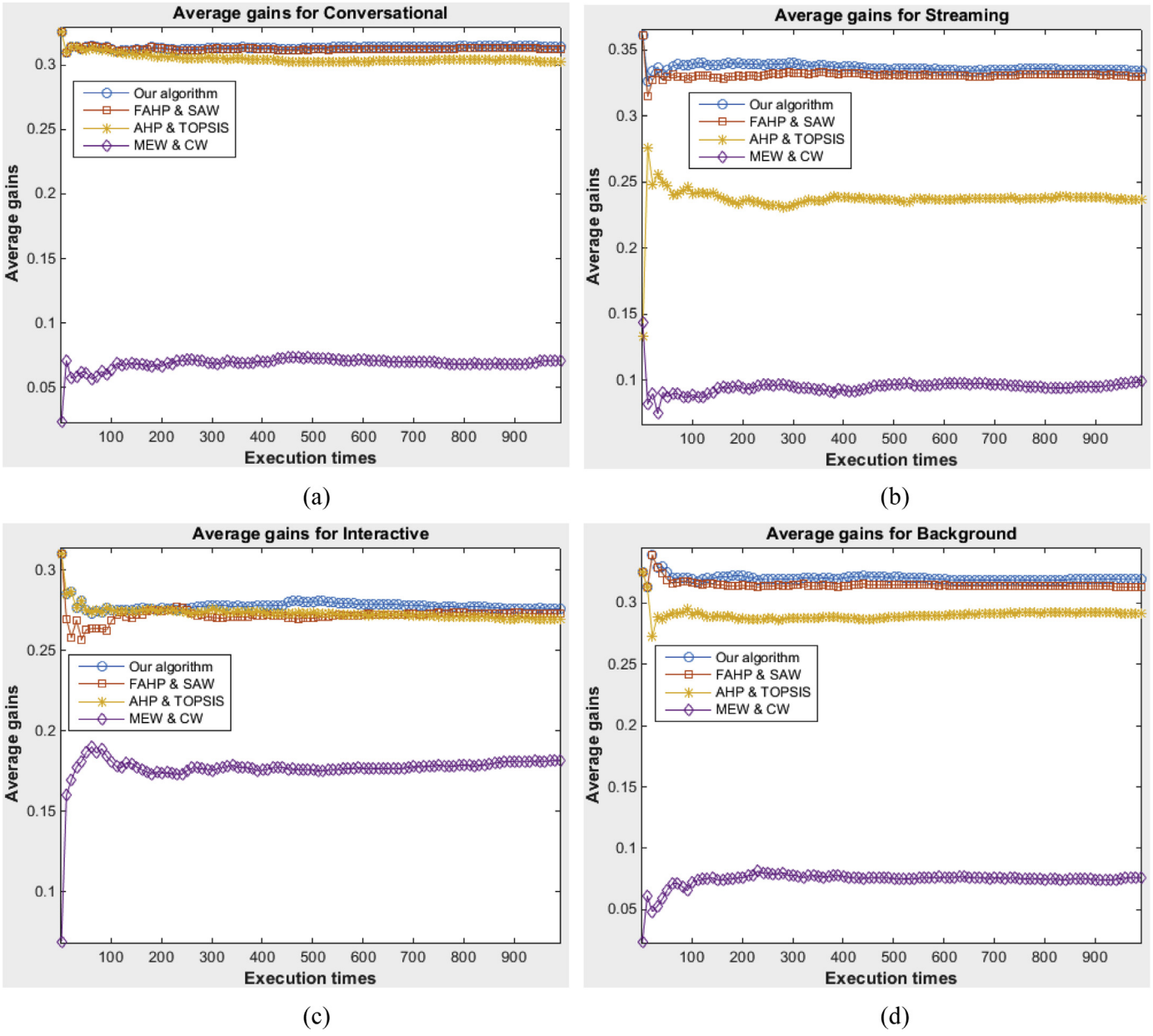


Fig. 7. Comparison of average values of gains for four traffic classes.

rithms, especially for the Algorithm 3. For all traffic classes, our algorithm and the Algorithm 1 show similar results and are much better than the Algorithm 3. In addition, for the traffic classes except conversational class, in the first 100 network selections, the average gain values of the algorithms except our algorithm are changed sharply. By contrast, our algorithm presents relatively stable values. This is because that once an algorithm is too sensitive to the environment, it has no ability to control potential ping-pong effect. This causes the UE to make frequent vertical handovers, and the corresponding average values of gains are low. The Algorithm 3 is a case in point. Hence, our algorithm effectively controls the number of vertical handovers, and provides the required QoS and better QoE.

5. Conclusions

This paper proposes an algorithm combined with improved FAHP, Entropy and TOPSIS to calculate the comprehensive utility values of candidate networks from two aspects of network attribute and user preference. The algorithm has fewer amounts of calculation by normalizing the decision matrix uniformly once for Entropy and TOPSIS. During the process of calculating the comprehensive utility values, it not only takes into account the subjective experience of the experts and users, and the basic service requirements of four kinds of traffic classes, but also considers the objective factor of the actual network environment. Therefore, it can get a relatively stable order of candidate networks. In addition, the values of adjustment coefficients can be flexibly adjusted according to specific traffic class or actual need. Moreover, with the change of these coefficients, the proportion of each algorithm in the final result can also be adjusted accordingly. Hence, it's the use of adjustment coefficients that makes the proposed scheme very flexible. Furthermore, the threshold δ is used to determine whether the UE should switch to the network with the largest utility value or continue to stay in the current one. This judgment mechanism can overcome the shortcomings of the traditional methods (i.e., $\delta = 1$),

which simply access the network with the largest utility value, and avoid some unnecessary vertical handoffs. According to the simulation results, the proposed algorithm not only selects the optimal target network, but also makes the total number of vertical handoffs significantly less than those of the three existing hybrid algorithms (FAHP & SAW, AHP & TOPSIS and MEW & CW) and provides the required QoS and QoE in terms of the quantified benefit from vertical handoff.

Finally, it should be noted that this paper has some limitations. On the one hand, the final adjustment coefficients (i.e. Table 14) and the threshold δ are gradually set through many experiments on the simulation scenario parameters (i.e. Table 13). Hence, there is no broad versatility. On the other hand, the proposed algorithm is to help a single mobile terminal to select the best network from the alternatives, while the load balancing of the alternatives has not been taken into account. These will be considered in our future research.

Acknowledgements

This study is supported by Guangdong Science & Technology Project (No. 2013B010401006), and Guangzhou Science & Technology Project (No. 2014J4100019).

References

[1] A. Ahmed, L.M. Boulahia, D. Gaiti, Enabling vertical handover decisions in heterogeneous wireless networks: a state-of-the-art and a classification, IEEE Commun. Surv. Tut. 16 (2) (2014) 776–811.
[2] S. Deswal, A. Singhrova, Handover algorithm for heterogeneous networks, in: 2016 3rd International Conference on Computing for Sustainable Global Development (INDIACom), New Delhi, 2016, pp. 3358–3364.
[3] T. Qiu, R. Qiao, M. Han, A.K. Sangaiah, I. Lee, A lifetime-enhanced data collecting scheme for internet of things, IEEE Commun. Mag. 55 (11) (2017) 132–137.
[4] T. Qiu, R. Qiao, D.O. Wu, EABS: an event-aware backpressure scheduling scheme for emergency internet of things, IEEE Trans. Mobile Comput. (2017), doi:10.1109/TMC.2017.2702670.
[5] T. Qiu, A. Zhao, F. Xia, W. Si, D.O. Wu, ROSE: robustness strategy for scale-free wireless sensor networks, IEEE/ACM Trans. Networking 25 (5) (2017) 2944–2959.
[6] Z. Ning, F. Xia, N. Ullah, X. Kong, X. Hu, Vehicular social networks: enabling smart mobility, IEEE Commun. Mag. 25 (5) (2017) 49–55.
[7] Z. Ning, X. Hu, Z. Chen, M. Zhou, B. Hu, J. Cheng, M. Obaidat, A cooperative quality-aware service access system for social internet of vehicles, IEEE Internet Things Journal (2017), doi:10.1109/JIOT.2017.2764259.
[8] C. Amali, D. Jayaprakash, B. Ramachandran, Enhanced media independent network selection for heterogeneous wireless networks, IETE Tech. Rev. 31 (5) (2014) 392–401.
[9] M. El Helou, S. Lahoud, M. Ibrahim, K. Khawam, B. Cousin, D. Mezher, A hybrid approach for radio access technology selection in heterogeneous wireless networks, Wireless Pers. Commun. 86 (2) (2016) 789–834.
[10] J.J. Escudero-Garzás, C. Bousoño-Calzón, An analysis of the network selection problem for heterogeneous environments with user-operator joint satisfactionMulti-RAT transmission, Wireless Commun. Mobile Comput. (2017) 1–13 Article ID 7425412, 2017, doi:10.1155/2017/7425412.
[11] M. Ali Hassoune, Z. Mekkakia Maaza, S.M. Senouci, Vertical handover decision algorithm for multimedia streaming in VANET, Wireless Pers. Commun. 95 (4) (2017) 4281–4299.
[12] R. Trestian, O. Ormond, G.M. Muntean, Performance evaluation of MADM-based methods for network selection in a multimedia wireless environment, Wireless Networks 21 (5) (2015) 1745–1763.
[13] A. Al-Rubaye, Y. Yeryomin, J. Seitz, Evaluation of MCDM-based handover decision algorithms in OMNeT++, in: 2016 3rd International Symposium on Wireless Systems within the Conferences on Intelligent Data Acquisition and Advanced Computing Systems (IDAACS-SWS), Offenburg, 2016, pp. 11–14.
[14] V. Raju, R,K. Garg, A.K. Dahiya, J. Kuriakose, A review on host vs. Network Mobility (NEMO) handoff techniques in heterogeneous network, in: 2014 3rd International Conference on Reliability, Infocom Technologies and Optimization (ICRITO) (Trends and Future Directions), Noida, 2014, pp. 1–5.
[15] O.A. Taiwo, O.E. Falowo, Comparative analysis of algorithms for making multiple-sessions handover decisions in next generation wireless networks, in: AFRICON, Pointe-Aux-Piments, 2013, pp. 1–6.
[16] B. Ma, H. Deng, X.Z. Xie, X.F. Liao, An optimized vertical handoff algorithm based on Markov process in vehicle heterogeneous network, China Commun. 12 (4) (2015) 106–116.
[17] D. Maroua, O. Mohammed, A. Driss, VIKOR for multi-criteria network selection in heterogeneous wireless networks, in: 2016 International Conference on Wireless Networks and Mobile Communications (WINCOM), Fez, 2016, pp. 82–86.
[18] I. Bisio, C. Braccini, S. Delucchi, F. Lavagetto, M. Marchese, Dynamic multi-attribute Network Selection algorithm for Vertical Handover procedures over mobile ad hoc networks, in: 2014 IEEE International Conference on Communications (ICC), Sydney, 2014, pp. 342–347.
[19] F.C. Yuan, T.Y. Chung, Practical voice session selection with FAHP in next generation wireless network, in: 2015 Seventh International Conference on Ubiquitous and Future Networks (ICUFN), Sapporo, 2015, pp. 196–201.
[20] E. Obayiuwana, O.E. Falowo, Network selection in heterogeneous wireless networks using multi-criteria decision-making algorithms: a review, Wireless Networks 23 (8) (2016) 2617–2649.
[21] O. Mämmelä, F.Z. Yousaf, P. Mannersalo, J. Lessmann, Policy-based and QoE-aware content delivery using Q-learning method, Wireless Pers. Commun. 83 (1) (2015) 315–342.
[22] P. Naghavi, S. Hamed Rastegar, V. Shah-Mansouri, H. Kebriaei, Learning RAT selection game in 5G heterogeneous networks, IEEE Wireless Commun. Lett. 5 (1) (2016) 52–55.
[23] S. Goudarzi, W.H. Hassan, M.H. Anisi, S.A. Soleymani, Comparison between hybridized algorithm of GA–SA and ABC, GA, DE and PSO for vertical-handover in heterogeneous wireless networks, in: Sādhanā-Academy Proceedings in Engineering Sciences, 41, 2016, pp. 727–753.
[24] B.R. Chandavarkar, R.M.R. Guddeti, Simplified and improved multiple attributes alternate ranking method for vertical handover decision in heterogeneous wireless networks, Computer Commun. 83 (2016) 81–97.
[25] R. Verma, N.P. Singh, GRA based network selection in heterogeneous wireless networks, Wireless Pers. Commun. 72 (2) (2013) 1437–1452.
[26] A. Habbal, S.I. Goudar, S. Hassan, Context-aware radio access technology selection in 5G ultra dense networks, IEEE Access 5 (2017) 6636–6648.
[27] B. Liu, H. Tian, B. Wang, B. Fan, AHP and game theory based approach for network selection in heterogeneous wireless networks, in: 2014 IEEE 11th Consumer Communications and Networking Conference (CCNC), Las Vegas, NV, 2014, pp. 501–506.
[28] M. Khan, K. Han, A vertical handover management scheme based on decision modelling in heterogeneous wireless networks, IETE Tech. Rev. 32 (6) (2015) 402–412.
[29] J. Zhou, C.Y. Zhu, Compensatory analysis and optimization for MADM for heterogeneous wireless network selection, J. Electr. Comput. Eng. 2016 (2016) 1–9.
[30] M. Fouladian, F. Hendessi, M.A. Pourmina, Using AHP and interval VIKOR methods to gateway selection in integrated VANET and 3G heterogeneous wireless networks in sparse situations, Arab. J. Sci. Eng. 41 (8) (2016) 2787–2800.
[31] V. Sasirekha, M. Ilanzkumaran, Heterogeneous wireless network selection using FAHP integrated with TOPSIS and VIKOR, in: 2013 International Conference on Pattern Recognition, Informatics and Mobile Engineering (PRIME), Salem, 2013, pp. 399–407.
[32] L. Li, X. Chu, J. Zhang, A hierarchical MADM-based network selection scheme for system performance enhancement, in: 2014 IEEE International Conference on Communication Systems (ICCS), Macau, 2014, pp. 122–126.
[33] D. Zhang, Y. Zhang, N. Lv, Y. He, An access selection algorithm based on GRA integrated with FAHP and entropy weight in hybrid wireless environment, in: 2013 7th International Conference on Application of Information and Communication Technologies (AICT), Baku, 2013, pp. 1–5.
[34] X.Q. Wang, Q. Guo, The priority of fuzzy complementary judgment matrix and its application in procurement tenders for government project, in: 2010 IEEE 17Th International Conference on Industrial Engineering and Engineering Management (IE&EM), Xiamen, 2010, pp. 148–151.
[35] E.M. Malathy, V. Muthuswamy, Knapsack - TOPSIS technique for vertical handover in heterogeneous wireless network, PLoS ONE 10 (8) (2015) 1–16.
[36] I. Chamodrakas, D. Martakos, A utility-based fuzzy TOPSIS method for energy efficient network selection in heterogeneous wireless networks, Appl. Soft Comput. 12 (7) (2012) 1929–1938.
[37] Z. Yue, A method for group decision-making based on determining weights of decision makers using TOPSIS, Appl. Math. Modell. 35 (4) (2011) 1926–1936.



He-Wei Yu is a professor of South China University of Technology. His research interests focus on integration technology of heterogeneous wireless networks. He directed several projects supported by the National Natural Science Foundation of China, Science & Technology projects of Guangdong Province, China.



Biao Zhang is a PhD candidate studying at South China University of Technology, advised by Dr. Yu. His research interests are novel algorithm for heterogeneous wireless networks selection, and resource allocation in next-generation wireless networks.